

(a) Baseline Architecture (YOLOv8s, 640×640 Resize) Duplicate rate: 25.4% | False Positives: 671 | Back-row F1: 0.1801

(b) Proposed TEMPO (1280×1280 Native + ByteTrack + Fog Recovery) Duplicate rate: 9.8% (-61.2%) | False Positives: 293 (-56.3%) | Back-row F1: 0.862

